

9A**AIM: Create and Explore a Dataset Using Pandas**

Create a student dataset containing Student ID, Name, Age, Gender, Course, and Marks. Store the data in a Pandas DataFrame and perform the following:

1. Display the complete dataset.
2. Display the first 5 records.
3. Display the last 5 records.
4. Display the number of rows and columns.
5. Display column names.
6. Display basic information about the dataset.
7. Display statistical information about numerical columns

CODE:

```
import pandas as pd
```

```
print("Shreyash Kadam S091")
```

```
data = {  
    "Student_ID": [101, 102, 103, 104, 105, 106, 107, 108],  
    "Name": ["Amit", "Sneha", "Rahul", "Priya", "Neha", "Rohan", "Pooja", "Karan"],  
    "Age": [19, 20, 19, 21, 20, 19, 21, 20],  
    "Gender": ["Male", "Female", "Male", "Female",  
               "Female", "Male", "Female", "Male"],  
    "Course": ["BSc CS", "BSc CS", "BSc IT", "BSc CS",  
               "BSc IT", "BSc CS", "BSc IT", "BSc CS"],  
    "Marks": [78, 85, 67, 92, 88, 74, 81, 69]  
}
```

```
df = pd.DataFrame(data)
```

```
print("Complete Dataset:")  
print(df)
```

```
print("\nFirst 5 Records:")  
print(df.head())
```

```
print("\nLast 5 Records:")  
print(df.tail())
```

```
print("\nShape:")  
print(df.shape)
```

```
print("\nColumn Names:")
```

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```
print(df.columns)
```

```
print("\nDataset Information:")
```

```
df.info()
```

```
print("\nStatistical Information:")
```

```
print(df.describe())
```

OUTPUT:

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Complete Dataset:

	Student_ID	Name	Age	Gender	Course	Marks
0	101	Amit	19	Male	BSc CS	78
1	102	Sneha	20	Female	BSc CS	85
2	103	Rahul	19	Male	BSc IT	67
3	104	Priya	21	Female	BSc CS	92
4	105	Neha	20	Female	BSc IT	88
5	106	Rohan	19	Male	BSc CS	74
6	107	Pooja	21	Female	BSc IT	81
7	108	Karan	20	Male	BSc CS	69

First 5 Records:

	Student_ID	Name	Age	Gender	Course	Marks
0	101	Amit	19	Male	BSc CS	78
1	102	Sneha	20	Female	BSc CS	85
2	103	Rahul	19	Male	BSc IT	67
3	104	Priya	21	Female	BSc CS	92
4	105	Neha	20	Female	BSc IT	88

Last 5 Records:

	Student_ID	Name	Age	Gender	Course	Marks
3	104	Priya	21	Female	BSc CS	92
4	105	Neha	20	Female	BSc IT	88
5	106	Rohan	19	Male	BSc CS	74
6	107	Pooja	21	Female	BSc IT	81
7	108	Karan	20	Male	BSc CS	69

Shape:

(8, 6)

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```
Column Names:
Index(['Student_ID', 'Name', 'Age', 'Gender', 'Course', 'Marks'], dtype='str')

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 8 entries, 0 to 7
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ------  -
0   Student_ID  8 non-null     int64
1   Name        8 non-null     str
2   Age         8 non-null     int64
3   Gender      8 non-null     str
4   Course      8 non-null     str
5   Marks       8 non-null     int64
dtypes: int64(3), str(3)
memory usage: 516.0 bytes

Statistical Information:
      Student_ID      Age      Marks
count      8.000000      8.000000      8.000000
mean     104.500000     19.875000     79.250000
std        2.44949      0.834523      8.940278
min       101.00000     19.000000     67.000000
25%       102.75000     19.000000     72.750000
50%       104.50000     20.000000     79.500000
75%       106.25000     20.250000     85.750000
max       108.00000     21.000000     92.000000
```

9B

AIM: Read and Explore a CSV Dataset

- Song_ID
- Song_Name
- Artist
- Genre
- Language
- Streams
- Rating

Write a Python program to:

1. Read the CSV file using Pandas.
2. Display the dataset.
3. Display the first 10 records.
4. Find the number of songs.
5. Display the column names.
6. Find the average rating.
7. Find the average number of streams.

Code:

```
import pandas as pd
```

```
print("Shreyash Kadam S091")
```

```
df = pd.read_csv("songs.csv")
```

```
print("Song Dataset:")  
print(df)
```

```
print("\nFirst 10 Records:")  
print(df.head(10))
```

```
print("\nNumber of Songs:")  
print(len(df))
```

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```
print("\nColumn Names:")  
print(df.columns)
```

```
print("\nAverage Rating:")  
print(df["Rating"].mean())
```

```
print("\nAverage Streams:")  
print(df["Streams"].mean())
```

Output:

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----- RESIARI: C:/Users/ASUS/OneDrive/Documents/Apps/Attachments/DESKTOP/Shreyash SICS SEM 3/

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Song Dataset:

	Song_ID	Song_Name	Artist	Genre	Language	Streams	Rating
0	201	Midnight Drive	Arjun Vale	Pop	English	1850000	8.4
1	202	Golden Hour	Mira Sen	Indie	Hindi	2640000	9.1
2	203	Ocean Eyes	Aarav Roy	Pop	English	3210000	8.7
3	204	Afterglow	Nisha K	Rock	English	1480000	7.9
4	205	Lost Stars	Rohan Mehta	Indie	Hindi	980000	7.5
5	206	Electric Heart	Tara Blue	EDM	English	4520000	9.3
6	207	Moonlight	Kabir Singh	Romantic	Hindi	1750000	8.1
7	208	Dreamscape	Meera Joshi	Pop	English	3890000	8.9
8	209	City Lights	Dev Arora	Hip Hop	Hindi	2180000	8.0
9	210	Echoes	Ishaan Ray	Rock	English	1260000	7.6
10	211	Rainy Letters	Anaya Verma	Romantic	Hindi	2950000	8.8
11	212	Neon Skies	Vihaan Rao	EDM	English	5170000	9.5
12	213	Paper Planes	Sana Malik	Indie	English	1120000	7.8
13	214	Slow Motion	Reyansh Shah	Pop	Hindi	3470000	8.6
14	215	Wildfire	Kiara Das	Rock	English	2030000	8.2
15	216	Heartbeat	Advik Jain	Romantic	Hindi	2780000	8.5
16	217	Street Dreams	Yuvaan Khan	Hip Hop	Hindi	4310000	9.0
17	218	Silver Moon	Riya Kapoor	Indie	English	890000	7.4
18	219	Gravity	Nakul Roy	EDM	English	3640000	8.8
19	220	Homebound	Ira Sen	Romantic	Hindi	1560000	8.3

First 10 Records:

	Song_ID	Song_Name	Artist	Genre	Language	Streams	Rating
0	201	Midnight Drive	Arjun Vale	Pop	English	1850000	8.4
1	202	Golden Hour	Mira Sen	Indie	Hindi	2640000	9.1
2	203	Ocean Eyes	Aarav Roy	Pop	English	3210000	8.7
3	204	Afterglow	Nisha K	Rock	English	1480000	7.9
4	205	Lost Stars	Rohan Mehta	Indie	Hindi	980000	7.5
5	206	Electric Heart	Tara Blue	EDM	English	4520000	9.3
6	207	Moonlight	Kabir Singh	Romantic	Hindi	1750000	8.1
7	208	Dreamscape	Meera Joshi	Pop	English	3890000	8.9
8	209	City Lights	Dev Arora	Hip Hop	Hindi	2180000	8.0
9	210	Echoes	Ishaan Ray	Rock	English	1260000	7.6

Number of Songs:

20

Number of Songs:

20

Column Names:

```
Index(['Song_ID', 'Song_Name', 'Artist', 'Genre', 'Language', 'Streams',
      'Rating'],
      dtype='str')
```

Average Rating:

8.420000000000002

Average Streams:

2584000.0

9C Selecting and Filtering Song Data**AIM:**

Using the `songs.csv` dataset, perform the following:

1. Display only Song Name and Rating.
2. Display songs that have a rating greater than 8.5.
3. Display songs having more than 3,000,000 streams.
4. Display songs belonging to the Pop genre.
5. Display songs available in Hindi language.
6. Display songs that have a rating greater than 8.5 and more than 3,000,000 streams.

CODE:

```
import pandas as pd
```

```
print("Shreyash Kadam S091")
```

```
df = pd.read_csv("songs.csv")
```

```
print("Song Name and Rating:")  
print(df[["Song_Name", "Rating"]])
```

```
print("\nSongs with Rating Greater Than 8.5:")  
print(df[df["Rating"] > 8.5])
```

```
print("\nSongs with More Than 3,000,000 Streams:")  
print(df[df["Streams"] > 3000000])
```

```
print("\nPop Songs:")  
print(df[df["Genre"] == "Pop"])
```

```
print("\nHindi Songs:")  
print(df[df["Language"] == "Hindi"])
```

```
print("\nSongs with Rating Greater Than 8.5 and Streams Above 3,000,000:")  
print(df[(df["Rating"] > 8.5) & (df["Streams"] > 3000000)])
```

OUTPUT:

```
Shreyash Kadam  
S091
```

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===== RESTART: C:/Users/Asus/OneDrive/Documents/Apps/Attachments/Desktop/Shreyash S091

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Song Name and Rating:

	Song_Name	Rating
0	Midnight Drive	8.4
1	Golden Hour	9.1
2	Ocean Eyes	8.7
3	Afterglow	7.9
4	Lost Stars	7.5
5	Electric Heart	9.3
6	Moonlight	8.1
7	Dreamscape	8.9
8	City Lights	8.0
9	Echoes	7.6
10	Rainy Letters	8.8
11	Neon Skies	9.5
12	Paper Planes	7.8
13	Slow Motion	8.6
14	Wildfire	8.2
15	Heartbeat	8.5
16	Street Dreams	9.0
17	Silver Moon	7.4
18	Gravity	8.8
19	Homebound	8.3

Songs with Rating Greater Than 8.5:

	Song_ID	Song_Name	Artist	Genre	Language	Streams	Rating
1	202	Golden Hour	Mira Sen	Indie	Hindi	2640000	9.1
2	203	Ocean Eyes	Aarav Roy	Pop	English	3210000	8.7
5	206	Electric Heart	Tara Blue	EDM	English	4520000	9.3
7	208	Dreamscape	Meera Joshi	Pop	English	3890000	8.9
10	211	Rainy Letters	Anaya Verma	Romantic	Hindi	2950000	8.8
11	212	Neon Skies	Vihaan Rao	EDM	English	5170000	9.5
13	214	Slow Motion	Reyansh Shah	Pop	Hindi	3470000	8.6
16	217	Street Dreams	Yuvaan Khan	Hip Hop	Hindi	4310000	9.0
18	219	Gravity	Nakul Roy	EDM	English	3640000	8.8

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Songs with More Than 3,000,000 Streams:

	Song_ID	Song_Name	Artist	Genre	Language	Streams	Rating
2	203	Ocean Eyes	Aarav Roy	Pop	English	3210000	8.7
5	206	Electric Heart	Tara Blue	EDM	English	4520000	9.3
7	208	Dreamscape	Meera Joshi	Pop	English	3890000	8.9
11	212	Neon Skies	Vihaan Rao	EDM	English	5170000	9.5
13	214	Slow Motion	Reyansh Shah	Pop	Hindi	3470000	8.6
16	217	Street Dreams	Yuvaan Khan	Hip Hop	Hindi	4310000	9.0
18	219	Gravity	Nakul Roy	EDM	English	3640000	8.8

Pop Songs:

	Song_ID	Song_Name	Artist	Genre	Language	Streams	Rating
0	201	Midnight Drive	Arjun Vale	Pop	English	1850000	8.4
2	203	Ocean Eyes	Aarav Roy	Pop	English	3210000	8.7
7	208	Dreamscape	Meera Joshi	Pop	English	3890000	8.9
13	214	Slow Motion	Reyansh Shah	Pop	Hindi	3470000	8.6

Hindi Songs:

	Song_ID	Song_Name	Artist	Genre	Language	Streams	Rating
1	202	Golden Hour	Mira Sen	Indie	Hindi	2640000	9.1
4	205	Lost Stars	Rohan Mehta	Indie	Hindi	980000	7.5
6	207	Moonlight	Kabir Singh	Romantic	Hindi	1750000	8.1
8	209	City Lights	Dev Arora	Hip Hop	Hindi	2180000	8.0
10	211	Rainy Letters	Anaya Verma	Romantic	Hindi	2950000	8.8
13	214	Slow Motion	Reyansh Shah	Pop	Hindi	3470000	8.6
15	216	Heartbeat	Advik Jain	Romantic	Hindi	2780000	8.5
16	217	Street Dreams	Yuvaan Khan	Hip Hop	Hindi	4310000	9.0
19	220	Homebound	Ira Sen	Romantic	Hindi	1560000	8.3

Songs with Rating Greater Than 8.5 and Streams Above 3,000,000:

	Song_ID	Song_Name	Artist	Genre	Language	Streams	Rating
2	203	Ocean Eyes	Aarav Roy	Pop	English	3210000	8.7
5	206	Electric Heart	Tara Blue	EDM	English	4520000	9.3
7	208	Dreamscape	Meera Joshi	Pop	English	3890000	8.9
11	212	Neon Skies	Vihaan Rao	EDM	English	5170000	9.5
13	214	Slow Motion	Reyansh Shah	Pop	Hindi	3470000	8.6
16	217	Street Dreams	Yuvaan Khan	Hip Hop	Hindi	4310000	9.0
18	219	Gravity	Nakul Roy	EDM	English	3640000	8.8

9D Sorting and Ranking Song Dataset**AIM:**

Using the song dataset:

1. Sort songs according to rating in ascending order.
2. Sort songs according to rating in descending order.
3. Sort songs according to streams in descending order.
4. Display the top 5 songs based on rating.
5. Display the bottom 3 songs based on rating

CODE:

```
import pandas as pd
```

```
df = pd.read_csv("songs.csv")
```

```
print("Rating in Ascending Order:")  
print(df.sort_values("Rating"))
```

```
print("\nRating in Descending Order:")  
print(df.sort_values("Rating", ascending=False))
```

```
print("\nStreams in Descending Order:")  
print(df.sort_values("Streams", ascending=False))
```

```
print("\nTop 5 Songs:")  
print(df.sort_values("Rating", ascending=False).head(5))
```

```
print("\nBottom 3 Songs:")  
print(df.sort_values("Rating").head(3))
```

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OUTPUT:

```
----- RESIDENT: C:/Users/ASUS/OneDrive/Documents/Apps/Attachments/Desktop/Shreyash S
Shreyash Kadam S091
Rating in Ascending Order:
  Song_ID      Song_Name      Artist      Genre Language Streams  Rating
17      218      Silver Moon    Riya Kapoor    Indie  English   890000    7.4
4       205      Lost Stars     Rohan Mehta     Indie  Hindi     980000    7.5
9       210      Echoes         Ishaan Ray      Rock   English  1260000    7.6
12      213      Paper Planes    Sana Malik      Indie  English  1120000    7.8
3       204      Afterglow       Nisha K         Rock   English  1480000    7.9
8       209      City Lights     Dev Arora       Hip Hop Hindi    2180000    8.0
6       207      Moonlight       Kabir Singh     Romantic Hindi    1750000    8.1
14      215      Wildfire        Kiara Das       Rock   English  2030000    8.2
19      220      Homebound       Ira Sen         Romantic Hindi    1560000    8.3
0       201      Midnight Drive  Arjun Vale      Pop    English  1850000    8.4
15      216      Heartbeat       Advik Jain      Romantic Hindi    2780000    8.5
13      214      Slow Motion     Reyansh Shah    Pop    Hindi    3470000    8.6
2       203      Ocean Eyes      Aarav Roy       Pop    English  3210000    8.7
18      219      Gravity         Nakul Roy       EDM    English  3640000    8.8
10      211      Rainy Letters   Anaya Verma     Romantic Hindi    2950000    8.8
7       208      Dreamscape      Meera Joshi     Pop    English  3890000    8.9
16      217      Street Dreams   Yuvaan Khan     Hip Hop Hindi    4310000    9.0
1       202      Golden Hour     Mira Sen        Indie  Hindi    2640000    9.1
5       206      Electric Heart  Tara Blue       EDM    English  4520000    9.3
11      212      Neon Skies      Vihaan Rao      EDM    English  5170000    9.5
```

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Rating in Descending Order:

	Song_ID	Song_Name	Artist	Genre	Language	Streams	Rating
11	212	Neon Skies	Vihaan Rao	EDM	English	5170000	9.5
5	206	Electric Heart	Tara Blue	EDM	English	4520000	9.3
1	202	Golden Hour	Mira Sen	Indie	Hindi	2640000	9.1
16	217	Street Dreams	Yuvaan Khan	Hip Hop	Hindi	4310000	9.0
7	208	Dreamscape	Meera Joshi	Pop	English	3890000	8.9
18	219	Gravity	Nakul Roy	EDM	English	3640000	8.8
10	211	Rainy Letters	Anaya Verma	Romantic	Hindi	2950000	8.8
2	203	Ocean Eyes	Aarav Roy	Pop	English	3210000	8.7
13	214	Slow Motion	Reyansh Shah	Pop	Hindi	3470000	8.6
15	216	Heartbeat	Advik Jain	Romantic	Hindi	2780000	8.5
0	201	Midnight Drive	Arjun Vale	Pop	English	1850000	8.4
19	220	Homebound	Ira Sen	Romantic	Hindi	1560000	8.3
14	215	Wildfire	Kiara Das	Rock	English	2030000	8.2
6	207	Moonlight	Kabir Singh	Romantic	Hindi	1750000	8.1
8	209	City Lights	Dev Arora	Hip Hop	Hindi	2180000	8.0
3	204	Afterglow	Nisha K	Rock	English	1480000	7.9
12	213	Paper Planes	Sana Malik	Indie	English	1120000	7.8
9	210	Echoes	Ishaan Ray	Rock	English	1260000	7.6
4	205	Lost Stars	Rohan Mehta	Indie	Hindi	980000	7.5
17	218	Silver Moon	Riya Kapoor	Indie	English	890000	7.4

Streams in Descending Order:

	Song_ID	Song_Name	Artist	Genre	Language	Streams	Rating
11	212	Neon Skies	Vihaan Rao	EDM	English	5170000	9.5
5	206	Electric Heart	Tara Blue	EDM	English	4520000	9.3
16	217	Street Dreams	Yuvaan Khan	Hip Hop	Hindi	4310000	9.0
7	208	Dreamscape	Meera Joshi	Pop	English	3890000	8.9
18	219	Gravity	Nakul Roy	EDM	English	3640000	8.8
13	214	Slow Motion	Reyansh Shah	Pop	Hindi	3470000	8.6
2	203	Ocean Eyes	Aarav Roy	Pop	English	3210000	8.7
10	211	Rainy Letters	Anaya Verma	Romantic	Hindi	2950000	8.8
15	216	Heartbeat	Advik Jain	Romantic	Hindi	2780000	8.5
1	202	Golden Hour	Mira Sen	Indie	Hindi	2640000	9.1
8	209	City Lights	Dev Arora	Hip Hop	Hindi	2180000	8.0
14	215	Wildfire	Kiara Das	Rock	English	2030000	8.2
0	201	Midnight Drive	Arjun Vale	Pop	English	1850000	8.4
6	207	Moonlight	Kabir Singh	Romantic	Hindi	1750000	8.1
19	220	Homebound	Ira Sen	Romantic	Hindi	1560000	8.3
3	204	Afterglow	Nisha K	Rock	English	1480000	7.9
9	210	Echoes	Ishaan Ray	Rock	English	1260000	7.6
12	213	Paper Planes	Sana Malik	Indie	English	1120000	7.8
4	205	Lost Stars	Rohan Mehta	Indie	Hindi	980000	7.5
17	218	Silver Moon	Riya Kapoor	Indie	English	890000	7.4

Top 5 Songs:

	Song_ID	Song_Name	Artist	Genre	Language	Streams	Rating
11	212	Neon Skies	Vihaan Rao	EDM	English	5170000	9.5
5	206	Electric Heart	Tara Blue	EDM	English	4520000	9.3
1	202	Golden Hour	Mira Sen	Indie	Hindi	2640000	9.1
16	217	Street Dreams	Yuvaan Khan	Hip Hop	Hindi	4310000	9.0
7	208	Dreamscape	Meera Joshi	Pop	English	3890000	8.9

Bottom 3 Songs:

	Song_ID	Song_Name	Artist	Genre	Language	Streams	Rating
17	218	Silver Moon	Riya Kapoor	Indie	English	890000	7.4
4	205	Lost Stars	Rohan Mehta	Indie	Hindi	980000	7.5
9	210	Echoes	Ishaan Ray	Rock	English	1260000	7.6

9E Handling Missing Values in Song Dataset**AIM:**

Consider a song dataset containing missing values in the Rating and Streams columns.

Perform the following:

1. Identify missing values.
2. Count missing values in each column.
3. Replace missing ratings with the average rating.
4. Replace missing streams with the average number of streams.
5. Display the cleaned dataset.

CODE:

```
import pandas as pd
```

```
data = {  
    "Song_Name": [  
        "Midnight Drive", "Golden Hour", "Ocean Eyes",  
        "Afterglow", "Lost Stars", "Electric Heart",  
        "Moonlight", "Dreamscape"  
    ],  
    "Genre": [  
        "Pop", "Indie", "Pop", "Rock",  
        "Indie", "EDM", "Romantic", "Pop"  
    ],  
    "Rating": [  
        8.4, None, 8.7, 7.9,  
        None, 9.3, 8.1, None  
    ],  
    "Streams": [  
        1850000, 2640000, None, 1480000,  
        980000, None, 1750000, 3890000  
    ]  
}
```

```
df = pd.DataFrame(data)
```

```
print("Original Dataset:")  
print(df)
```

```
print("\nMissing Values:")  
print(df.isnull())
```

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```
print("\nCount of Missing Values:")
```

```
print(df.isnull().sum())
```

```
df["Rating"] = df["Rating"].fillna(df["Rating"].mean())
```

```
df["Streams"] = df["Streams"].fillna(df["Streams"].mean())
```

```
print("\nCleaned Dataset:")
```

```
print(df)
```

OUTPUT:

```
===== RESTART: C:/Users/Asus/OneDrive/Documents/Apps/Attachments/Desktc
```

```
Original Dataset:
```

	Song_Name	Genre	Rating	Streams
0	Midnight Drive	Pop	8.4	1850000.0
1	Golden Hour	Indie	NaN	2640000.0
2	Ocean Eyes	Pop	8.7	NaN
3	Afterglow	Rock	7.9	1480000.0
4	Lost Stars	Indie	NaN	980000.0
5	Electric Heart	EDM	9.3	NaN
6	Moonlight	Romantic	8.1	1750000.0
7	Dreamscape	Pop	NaN	3890000.0

```
Missing Values:
```

	Song_Name	Genre	Rating	Streams
0	False	False	False	False
1	False	False	True	False
2	False	False	False	True
3	False	False	False	False
4	False	False	True	False
5	False	False	False	True
6	False	False	False	False
7	False	False	True	False

```
Count of Missing Values:
```

```
Song_Name    0
Genre        0
Rating       3
Streams      2
dtype: int64
```

```
Cleaned Dataset:
```

	Song_Name	Genre	Rating	Streams
0	Midnight Drive	Pop	8.40	1.850000e+06
1	Golden Hour	Indie	8.48	2.640000e+06
2	Ocean Eyes	Pop	8.70	2.098333e+06
3	Afterglow	Rock	7.90	1.480000e+06
4	Lost Stars	Indie	8.48	9.800000e+05
5	Electric Heart	EDM	9.30	2.098333e+06
6	Moonlight	Romantic	8.10	1.750000e+06
7	Dreamscape	Pop	8.48	3.890000e+06

9F Data Cleaning and Adding New Columns**AIM:**

Using the song dataset, create a new column called **Popularity**.

The song should be classified as:

- **"Highly Popular"** if Streams $\geq 3,000,000$
- **"Popular"** if Streams $\geq 2,000,000$
- **"Less Popular"** if Streams $< 2,000,000$

Also create a **Rating_Category** column using:

Rating	Category
9.0–10.0	Excellent
8.0–8.9	Very Good
7.0–7.9	Good
Below 7.0	Average

CODE:

```
import pandas as pd
```

```
df = pd.read_csv("songs.csv")
```

```
df["Popularity"] = df["Streams"].apply(  
    lambda x: "Highly Popular" if x >= 3000000 else "Popular" if x >= 2000000 else "Less  
Popular"  
)
```

```
def calculate_rating_category(rating):
```

```
    if rating >= 9:  
        return "Excellent"  
    elif rating >= 8:  
        return "Very Good"  
    elif rating >= 7:  
        return "Good"  
    else:  
        return "Average"
```

```
df["Rating_Category"] = df["Rating"].apply(calculate_rating_category)
```

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S091

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print(df)

OUTPUT:

```
===== RESTART: C:/Users/Asus/OneDrive/Documents/Apps/Attachments/Desktop/Shreyash S:
Shreyash Kadam S091
   Song_ID  Song_Name  ...  Popularity Rating_Category
0      201  Midnight Drive  ...  Less Popular  Very Good
1      202   Golden Hour  ...    Popular  Excellent
2      203   Ocean Eyes  ...  Highly Popular  Very Good
3      204   Afterglow  ...  Less Popular    Good
4      205   Lost Stars  ...  Less Popular    Good
5      206  Electric Heart  ...  Highly Popular  Excellent
6      207   Moonlight  ...  Less Popular  Very Good
7      208   Dreamscape  ...  Highly Popular  Very Good
8      209   City Lights  ...    Popular  Very Good
9      210    Echoes  ...  Less Popular    Good
10     211  Rainy Letters  ...    Popular  Very Good
11     212   Neon Skies  ...  Highly Popular  Excellent
12     213  Paper Planes  ...  Less Popular    Good
13     214   Slow Motion  ...  Highly Popular  Very Good
14     215   Wildfire  ...    Popular  Very Good
15     216   Heartbeat  ...    Popular  Very Good
16     217  Street Dreams  ...  Highly Popular  Excellent
17     218   Silver Moon  ...  Less Popular    Good
18     219   Gravity  ...  Highly Popular  Very Good
19     220   Homebound  ...  Less Popular  Very Good

[20 rows x 9 columns]
```

9G Statistical Analysis of Song Dataset**AIM:**

Using `songs.csv`, calculate:

1. Average rating.
2. Maximum rating.
3. Minimum rating.
4. Median rating.
5. Standard deviation of rating.
6. Average number of streams.
7. Number of songs.
8. Number of songs that have a rating greater than 8.5.

CODE:

```
import pandas as pd
```

```
df = pd.read_csv("songs.csv")
```

```
print("Average Rating:", df["Rating"].mean())
```

```
print("Maximum Rating:", df["Rating"].max())
```

```
print("Minimum Rating:", df["Rating"].min())
```

```
print("Median Rating:", df["Rating"].median())
```

```
print("Standard Deviation:", df["Rating"].std())
```

```
print("Average Streams:", df["Streams"].mean())
```

```
print("Number of Songs:", df["Song_Name"].count())
```

```
print("Songs with Rating Above 8.5:",  
      (df["Rating"] > 8.5).sum())
```

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OUTPUT:

```
Shreyash Kadam S091  
Average Rating: 8.420000000000002  
Maximum Rating: 9.5  
Minimum Rating: 7.4  
Median Rating: 8.45  
Standard Deviation: 0.6040216099306034  
Average Streams: 2584000.0  
Number of Songs: 20  
Songs with Rating Above 8.5: 9
```

9H Grouping and Aggregation of Song Dataset**AIM:**

Using the song dataset, perform analysis based on the **Genre** column:

1. Find the number of songs in each genre.
2. Find the average rating for each genre.
3. Find the maximum rating for each genre.
4. Find the average number of streams for each genre.

CODE:

```
import pandas as pd
```

```
df = pd.read_csv("songs.csv")
```

```
print("Number of Songs in Each Genre:")  
print(df.groupby("Genre")["Song_Name"].count())
```

```
print("\nAverage Rating by Genre:")  
print(df.groupby("Genre")["Rating"].mean())
```

```
print("\nMaximum Rating by Genre:")  
print(df.groupby("Genre")["Rating"].max())
```

```
print("\nAverage Streams by Genre:")  
print(df.groupby("Genre")["Streams"].mean())
```

OUTPUT:

```
===== RESTART: C:/Users/Asus/OneDrive  
Shreyash Kadam S091  
Number of Songs in Each Genre:  
Genre  
EDM          3  
Hip Hop      2  
Indie        4  
Pop          4  
Rock         3  
Romantic     4  
Name: Song_Name, dtype: int64  
  
Average Rating by Genre:  
Genre  
EDM          9.200  
Hip Hop      8.500  
Indie        7.950  
Pop          8.650  
Rock         7.900  
Romantic     8.425  
Name: Rating, dtype: float64
```

Maximum Rating by Genre:

Genre

EDM 9.5

Hip Hop 9.0

Indie 9.1

Pop 8.9

Rock 8.2

Romantic 8.8

Name: Rating, dtype: float64

Average Streams by Genre:

Genre

EDM 4.443333e+06

Hip Hop 3.245000e+06

Indie 1.407500e+06

Pop 3.105000e+06

Rock 1.590000e+06

Romantic 2.260000e+06

Name: Streams, dtype: float64

Number of Songs in Each Genre:

Genre

EDM 3

Hip Hop 2

Indie 4

Pop 4

Rock 3

Romantic 4

Name: Song_Name, dtype: int64

Average Rating by Genre:

Genre

EDM 9.200

Hip Hop 8.500

Indie 7.950

Pop 8.650

Rock 7.900

Romantic 8.425

Name: Rating, dtype: float64

Average Rating by Genre:

Genre

EDM 9.200

Hip Hop 8.500

Indie 7.950

Pop 8.650

Rock 7.900

Romantic 8.425

Name: Rating, dtype: float64

Maximum Rating by Genre:

Genre

EDM 9.5

Hip Hop 9.0

Indie 9.1

Pop 8.9

Rock 8.2

Romantic 8.8

Name: Rating, dtype: float64

Average Streams by Genre:

Genre

EDM 4.443333e+06

Hip Hop 3.245000e+06

Indie 1.407500e+06

Pop 3.105000e+06

Rock 1.590000e+06

Romantic 2.260000e+06

Name: Streams, dtype: float64

9I Data Visualization Using Matplotlib**AIM:**

Using the song dataset, create the following charts:

1. Bar chart showing song names and ratings.
2. Line chart showing song names and ratings.
3. Histogram showing the distribution of ratings.
4. Pie chart showing the number of songs in each genre.

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("songs.csv")

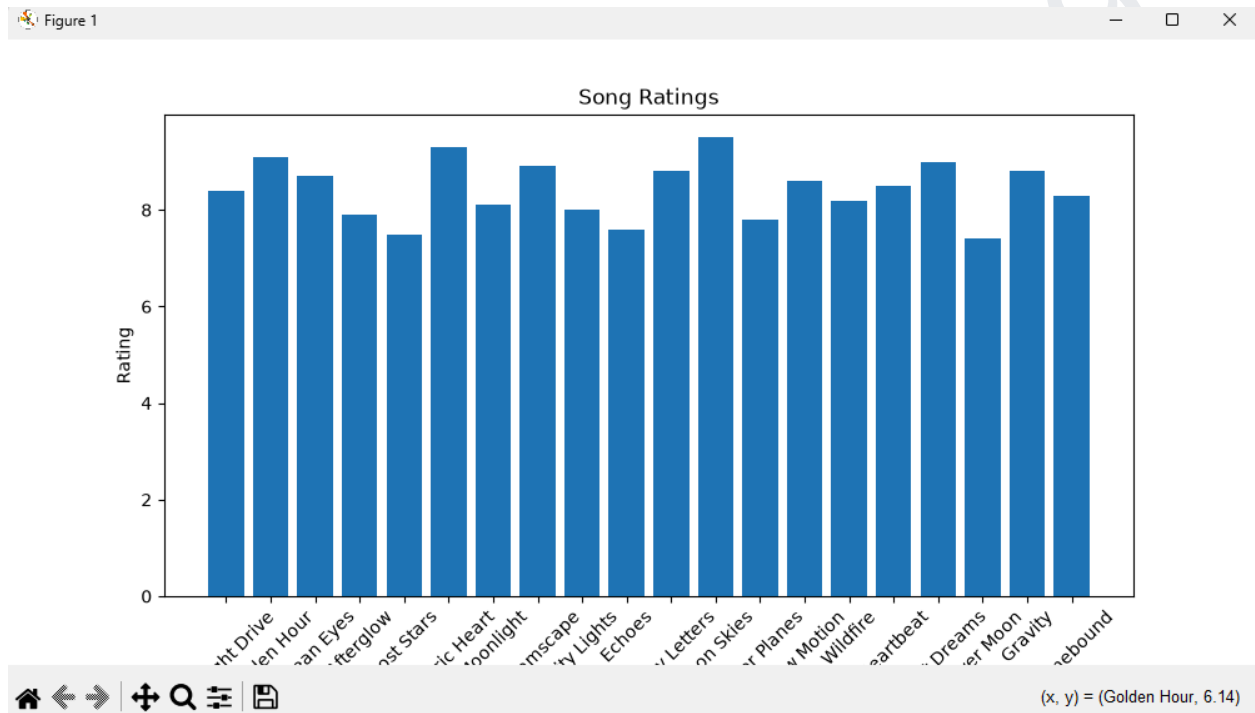
plt.figure(figsize=(10, 5))
plt.bar(df["Song_Name"], df["Rating"])
plt.xlabel("Song Name")
plt.ylabel("Rating")
plt.title("Song Ratings")
plt.xticks(rotation=45)
plt.show()

plt.figure(figsize=(10, 5))
plt.plot(df["Song_Name"], df["Rating"], marker="o")
plt.xlabel("Song Name")
plt.ylabel("Rating")
plt.title("Song Ratings - Line Chart")
plt.xticks(rotation=45)
plt.show()

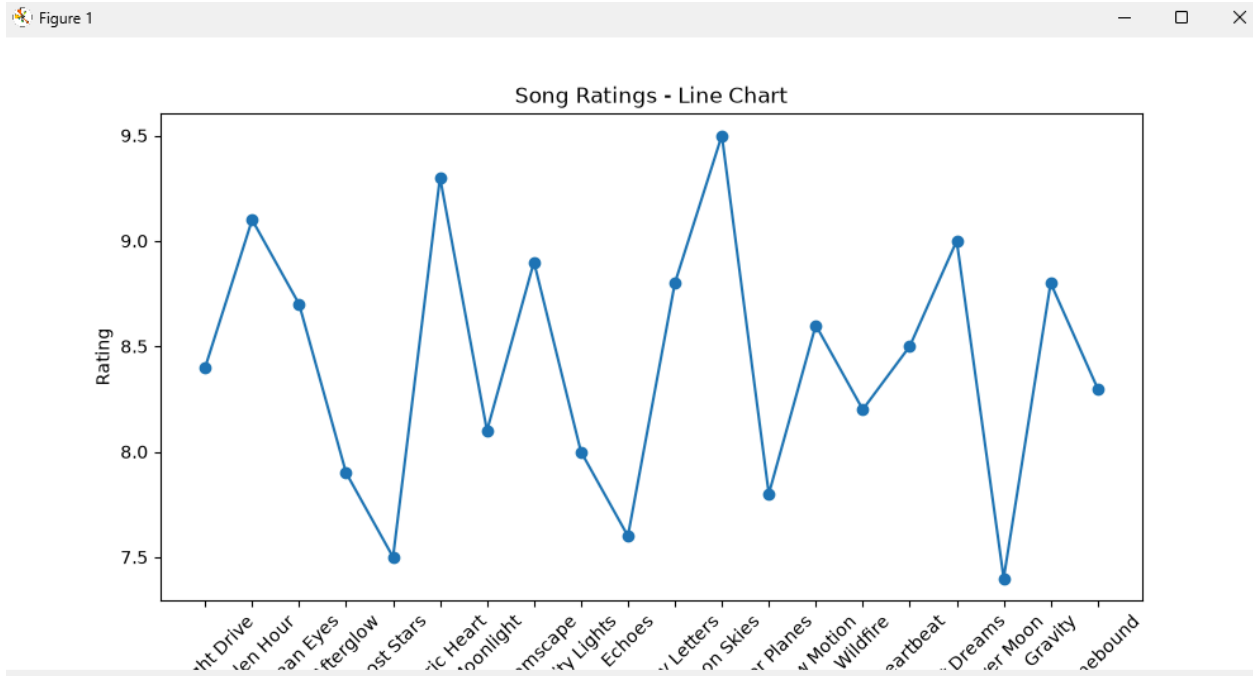
plt.figure(figsize=(8, 5))
plt.hist(df["Rating"], bins=5)
plt.xlabel("Rating")
plt.ylabel("Number of Songs")
plt.title("Distribution of Song Ratings")
plt.show()

genre_count = df["Genre"].value_counts()
```

```
plt.figure(figsize=(7, 7))
plt.pie(
    genre_count,
    labels=genre_count.index,
    autopct="%1.1f%%"
)
plt.title("Songs by Genre")
plt.show()
```

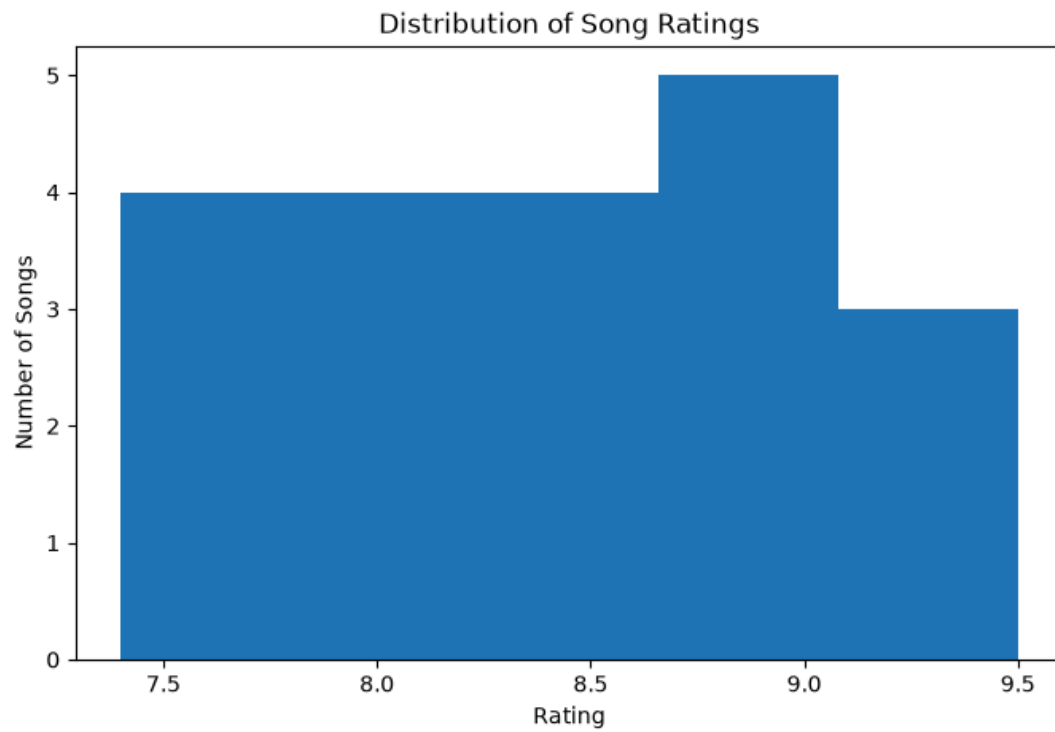
OUTPUT:

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Figure 1



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